Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

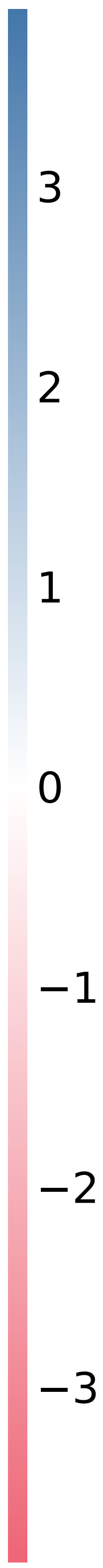
F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV



Reasoning + Answer:

<think>

To find the de Broglie wavelength, we use the formula Î» = h / (sqrt(2mE)), where h is Planck's constant, m is the mass of the electron, and E is the kinetic energy. We need to convert 1eV to joules for the calculation. 1eV = 1.602 ÃĹ 10^-19 J. The mass of an electron is 9.109 ÃĹ 10^-31 kg. Plugging in the values:

Î» = (6.626 ÃĹ 10^-34 J*s / (sqrt(2*9.109 ÃĹ 10^-31 kg * 1.602 ÃĹ 10^-19 J))) = 2.427 ÃĹ 10^-10 m âīĪ 2.48 ÃĹ 10^-8 cm

The closest answer to this calculation is 2.48 ÃĹ 10^-8 cm.

</think>

<answer>2.48 ÃĹ 10^-8 cm</answer>